

Uthanattu Sunil Amalkumar

Azure DevOps Engineer | DevSecOps | Terraform | AKS | CI/CD

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PROFESSIONAL SUMMARY

Azure DevOps Engineer with **6 years** of experience designing and operating scalable **CI/CD pipelines**, **Terraform**-based infrastructure, and **AKS** workloads on Microsoft Azure. Hands-on across **Azure Pipelines**, **Function Apps**, **Logic Apps**, **ADF**, networking (Application Gateway, Private DNS / Endpoints), and security and governance with **Entra ID**, **PIM**, custom RBAC, **Checkmarx**, and **SonarQube**. Proven track record of automating disaster recovery (ASR), backup, monitoring (**Prometheus**, **Grafana**, Azure Monitor), and cost optimization — accelerating delivery and reducing operational overhead.

TECHNICAL SKILLS

- **Cloud & Azure Services:** Microsoft Azure (IaaS, PaaS), Azure Virtual Machines, Function Apps, Logic Apps, Web Apps, Azure Data Factory (ADF), Service Bus, Key Vault, Storage Accounts, Azure Runbooks, Azure Update Manager, Azure Site Recovery (ASR), Azure Backup
- **CI/CD & Automation:** Azure DevOps (Pipelines, Repos, Artifacts, Boards), YAML Pipelines, Approval Gates, PowerShell, Bash Scripting
- **Infrastructure as Code:** Terraform (reusable modules, state management, resource imports, refactoring), ARM Templates
- **Containers & Kubernetes:** Docker, Azure Kubernetes Service (AKS), Azure Container Registry (ACR), Helm, Ingress Controllers, Network Policies, HPA, Cluster Autoscaler
- **Networking:** Virtual Networks (VNet), Subnets, NSG, ASG, Load Balancers, Application Gateway, Azure DNS, Private DNS, Private Endpoints
- **Security, Identity & Governance:** Entra ID (Azure AD), Privileged Identity Management (PIM), Custom RBAC Roles, Checkmarx (SAST), SonarQube, Quality Gates
- **Monitoring & Observability:** Prometheus, Grafana, Alertmanager, Azure Monitor, Log Analytics
- **Tools & Methodologies:** Git, Azure Repos, Agile, Scrum, DevOps, SRE

PROFESSIONAL EXPERIENCE

IBM | Hyderabad, India

Jan 2025 – Present

Azure DevOps Engineer

- Engineered reusable Terraform modules to provision Azure Virtual Machines with built-in domain joining, **reducing VM build time from 2–3 hours to under 15 minutes (~85% faster)** and eliminating manual post-build configuration steps.
- Architected Terraform modules to provision complete integration stacks (Function Apps, Logic Apps, App Service Plans, Key Vaults, Service Bus, ADF, Private Endpoints) standardized across **5 environments (Dev/SIT/UAT/Pre-Prod/Prod)**, **cutting provisioning from multi-day builds to under 20 minutes (~70% effort reduction)** and lowering idle infrastructure cost by **~30%**.
- Automated tag-based scheduled start/stop of non-production VMs using Azure Runbooks, **reducing monthly compute costs by ~40%** on dev/test workloads.
- Authored custom Azure RBAC roles and onboarded teams to Privileged Identity Management (PIM), **enforcing least-privilege access** and passing internal access-control audits with zero high-severity findings.
- Implemented end-to-end VM data protection — Azure Backup across **100% of production VMs** plus an automated snapshot lifecycle and cleanup strategy — ensuring point-in-time recovery, full data-protection compliance, and **reducing snapshot storage spend by ~60%**.
- Configured Azure Site Recovery (ASR) and executed Disaster Recovery drills for critical workloads, **achieving RPO <15 min and RTO <1 hour** to meet business continuity SLAs.
- Built end-to-end CI/CD pipelines in **Azure DevOps** to deploy Function Apps and Logic Apps code across multiple environments with gated approvals, integrating **SonarQube quality gates** (raising code coverage

to **80%+**) and **Checkmarx SAST scans** to detect and remediate vulnerabilities pre-merge, **reducing release cycle time from days to under 1 hour**, eliminating manual deployment errors, and significantly cutting production security findings.

- Designed ADF CI/CD pipelines to promote pipelines, datasets, and linked services across environments, **automating 100% of data-integration releases** previously done manually.

Tata Consultancy Services (TCS) | Hyderabad, India

Aug 2020 – Jan 2025

Cloud Engineer

- Built end-to-end CI/CD pipelines in Azure DevOps to automate containerized application deployments from Azure Container Registry (ACR) to Azure Kubernetes Service (AKS) with integrated code quality and security scans, **reducing deployment time from ~2 hours to under 15 minutes (~85% faster)** and removing manual release errors.
- Architected and operated production AKS clusters hosting **30+ microservices**, selecting the right workload primitives (Deployments, DaemonSets, StatefulSets) for stateless APIs, node-level agents, and stateful services to deliver high availability for internal and external consumers.
- Designed AKS auto-scaling using **Horizontal Pod Autoscaler (HPA)** and **Cluster Autoscaler** to dynamically right-size pods and nodes — **absorbing 3x peak-load traffic spikes with zero downtime, sustaining 99.9% application uptime, and lowering compute costs by ~35%** while preserving performance SLAs.
- Implemented AKS ingress with **NGINX** and **Application Gateway Ingress Controller (AGIC)** — TLS termination, path-based routing, and Azure DNS / Private DNS integration — **consolidating 10+ public endpoints** behind a single hardened ingress layer.
- Implemented end-to-end observability for AKS using **Prometheus**, **Grafana**, and **Alertmanager** — metrics collection, dashboards on node health, pod utilization, and application SLIs/SLOs, plus custom alert rules with on-call notifications — **reducing incident detection time by ~60% and Mean Time To Recovery (MTTR) by ~50%**.
- Managed core Azure infrastructure (Virtual Machines, Entra ID, Storage Accounts) and designed network segmentation with VNets, Subnets, NSGs, and ASGs — **eliminating flat-network risk** and aligning with enterprise security baselines across multiple application teams.

CERTIFICATIONS

- Microsoft Certified: Azure Solutions Architect Expert (AZ-305)
- Microsoft Certified: DevOps Engineer Expert (AZ-400)
- HashiCorp Certified: Terraform Associate (003)
- Microsoft Certified: Azure Administrator Associate (AZ-104)

EDUCATION

Bachelor of Technology (B.Tech)

2016 – 2020

JNTU College of Engineering, Pulivendula, India